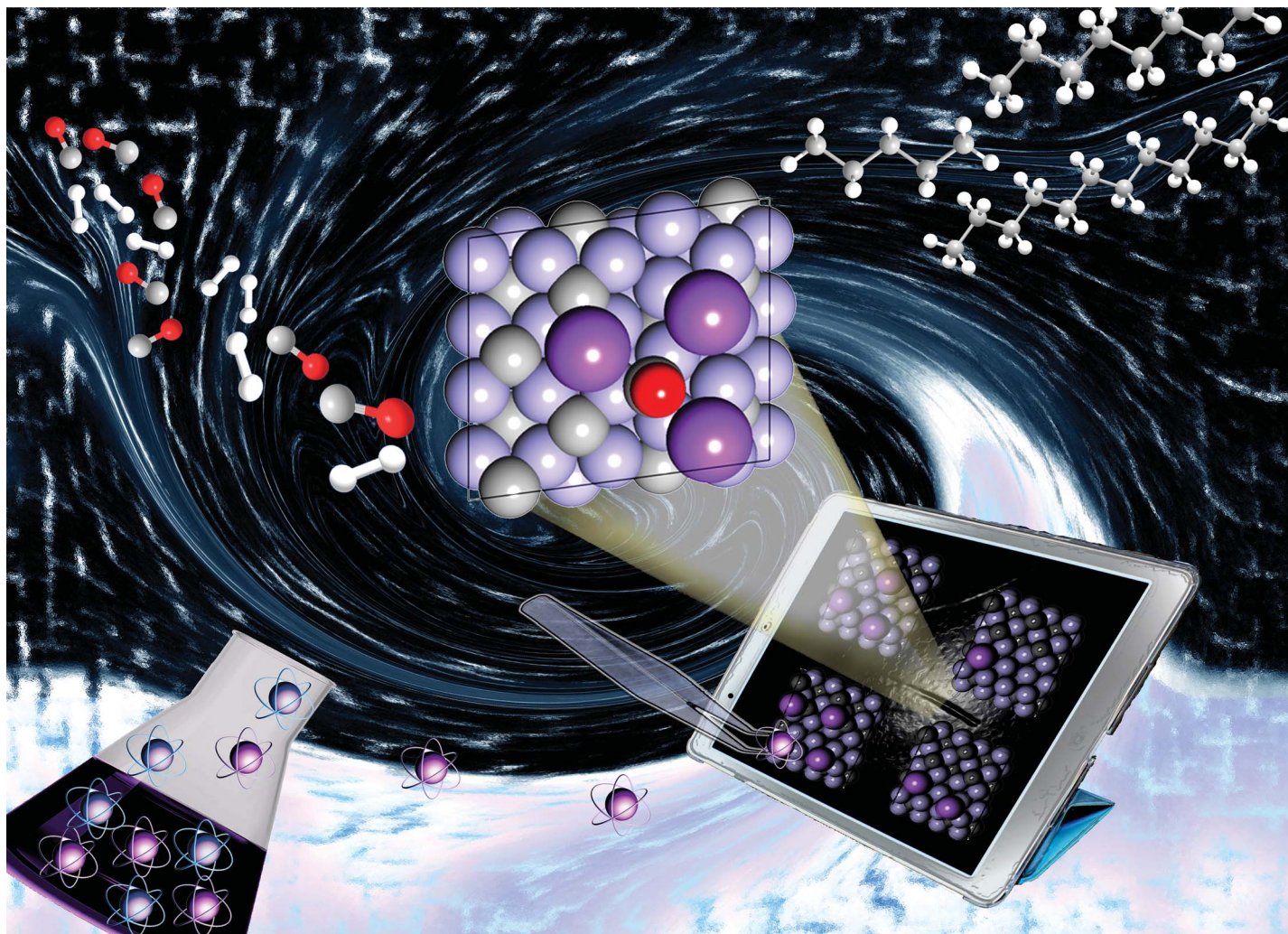


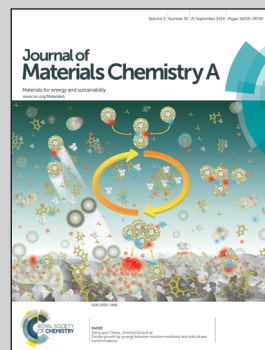


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Title: Highly activated K-doped iron carbide nanocatalysts designed by computational simulation for Fischer–Tropsch synthesis

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Highly activated K-doped iron carbide nanocatalysts designed by computational simulation for Fischer–Tropsch synthesis†

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Although the reaction results of numerous iron-based Fischer–Tropsch synthesis catalysts containing various promoters have been reported, the research on their theoretical foundation is still insufficient. In the present work, highly activated K-doped χ -Fe₅C₂/charcoal nanocatalysts were designed using calculations based on density functional theory (DFT), and then prepared using a melt-infiltration process and a subsequent incipient-wetness method of K precursors. The catalyst at K/Fe = 0.075 in an atomic ratio that bears small iron carbide nanoparticles of ~18 nm showed the highest activity (1.54×10^{-4} mol_{CO} g_{Fe}⁻¹ s⁻¹) and the best hydrocarbon yield (1.41×10^{-3} g_{HC} g_{Fe}⁻¹ s⁻¹), as well as a good selectivity for gasoline-range (C₅–C₁₂) hydrocarbon products in the high-temperature Fischer–Tropsch reaction.

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1. Introduction

Fischer–Tropsch synthesis (FTS) is known as an effective method for producing long straight-chain hydrocarbons, by a surface polymerization reaction, from a mixture of CO and H₂ gasses.^{1–4} In particular, the high-temperature FTS reaction, which is normally conducted at temperatures of 300–350 °C, has been a key technology for producing automotive gasoline components and lower olefins from fossil resources (e.g., natural gas, biomass, or coal).^{5–7}

In the preparation of the catalyst, the choice of porous support materials is crucial to guarantee the uniform dispersion of active metal particles and to protect from severe agglomeration of the catalyst particles. Until now, various Fe-supported catalysts using γ -Al₂O₃, SiO₂, and MgO have been applied to the FTS reaction.^{8–10} However, when these metal-oxide supports were used, a strong metal–support interaction was also generated, leading to difficulties in reduction and activation to catalytically active phase(s). For high-temperature reactions, porous carbon materials such as activated charcoal/carbon appear to be good candidates for catalyst support due to

their high chemical stability, good thermal conductivity, and weak metal–carbon interaction.^{11–15} Recently, Galvis *et al.* showed high activity using a Fe-on-carbon-nanofiber (CNF) catalyst as well as good selectivity for lower olefins in high temperature FTS.^{16,17}

Because iron-oxides (Fe₂O₃, Fe₃O₄), metallic iron, and iron carbides usually coexist during FTS, the exact determination of the real active phases is very difficult. Although the identity of the active phases remains controversial, the monoclinic Hägg-carbide (χ -Fe₅C₂) has been considered to be one of the most important active species among several iron, iron oxide, and iron carbide phases (e.g., ϵ -Fe₂C, ϵ' -Fe_{2.2}C, Fe₇C₃, θ -Fe₃C).¹⁸ Many studies on active χ -Fe₅C₂ formation have been conducted by controlling the activation condition of inactive iron-oxides, and activation with CO gas has generally led to higher initial activity than that with synthesis gas or H₂.^{19–23} In order to exclude the long extra activation step under CO or mixed gas, Yang *et al.* reported pure Hägg-carbide nanoparticles (around 20 nm), directly synthesized by the bromide-induced wet chemical method.²⁴ Besides adjusting an activation process, some additives, such as Cu, K, and Zn which affect the electronic and geometric properties of the catalyst surface, were widely exploited to enhance CO conversion and product selectivity.^{25–28} In particular, alkali metals have been considered to be excellent additives because they can easily donate electrons to active surfaces, thus enhancing the catalyst basicity.^{29,30} For instance, the addition of potassium (K) into an iron catalyst as a base promoter has dramatically affected catalytic behavior, even in extremely small amounts (K/Fe at 0.01–0.04 atomic ratios).^{31–33}

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† Electronic supplementary information (ESI) available: Hägg-carbide model of a 1 × 2 super-cell, TEM images and particle size distribution histograms of K-doped χ -Fe₅C₂ nanoparticles, Fe K-edge X-ray absorption spectra, and CO conversion and selectivity data for K-free and K-doped χ -Fe₅C₂/charcoal catalysts. See DOI: 10.1039/c4ta02413c

Recently, computational simulation based on density functional theory (DFT) of surface reactions has become an effective tool for the design of new catalysts.^{34–38} Along this line, several theoretical studies on iron carbide catalysts have involved using the computed activation energy, surface energy, and reaction energy.^{39–41} In recent years, Sorescu reported the adsorption and dissociation of CO on K-doped Fe(100) surfaces and showed that an increase of the surface area covered by K could lead to dramatically enhanced adsorption or dissociation behaviors.^{42,43} Huo *et al.* explained the enhanced activity of the K-promoted metallic Fe catalyst theoretically and experimentally, and proposed that the main reason for enlarged active facets was the controlled Fe-crystal growth rate induced by added K.⁴⁴ However, the optimum atomic ratio of K per Fe, especially in the χ -Fe₅C₂ structure for FTS, has not been clearly described in either theory or experiment, thus far.

Herein, we report a facile new method for the synthesis of highly activated K-doped χ -Fe₅C₂/charcoal catalysts *via* melt-infiltration of the iron salt and subsequent incipient wetness impregnation of K precursors. The design was based on DFT calculation aimed at achieving the K addition effect. The obtained K-doped χ -Fe₅C₂/charcoal catalyst (K/Fe atomic ratio = 0.075) showed the highest activity (1.54×10^{-4} mol_{CO} g_{Fe}⁻¹ s⁻¹), as well as a good selectivity for gasoline-range (C₅–C₁₂) hydrocarbon products in the high-temperature FTS reaction under 320 °C and at 15 bar. The resulting activity of the optimally K-doped χ -Fe₅C₂/charcoal catalyst is comparable to the highest value previously reported among various supported-iron nanocatalysts.¹⁶ In addition, this catalyst design based on DFT simulation and the simple synthesis method could remarkably reduce the optimization process by the conventional time-consuming trial-and-error method, for complicated catalysts with promoters.

2. Experimental section

Chemicals

Iron nitrate nonahydrate (Fe(NO₃)₃·9H₂O, ACS reagent, ≥98%), activated charcoal (~100 mesh particle size, powder), K₂CO₃ (powder, reagent grade, ≥98%), glass beads (425–600 μm size), and anhydrous ethanol were purchased from Aldrich. The chemicals were used as received without further purification.

Computations

DFT calculations were performed using the Vienna *ab initio* simulation package (VASP). This plane-wave code employs a projected, augmented wave with a generalized gradient approximation (PAW-GGA) for the exchange–correlation functional using the PBE method.⁴⁵ We performed geometric optimizations by minimizing the total energy of the unit cell using a conjugated-gradient algorithm to relax the ions. The Fe₅C₂ phase has a monoclinic structure and contains 20 Fe and 8 C atoms per unit cell. The corresponding optimized lattice parameters are in good agreement with the results of previous studies: $a = 11.580$ Å, $b = 4.508$ Å, $c = 4.994$ Å, and $\beta = 97.64^\circ$.⁴⁶ For the Fe₅C₂ surface, we focused on the stable surface structure

to estimate and characterize the reactivity of the Fe₅C₂ surfaces in the presence of K. The Fe₅C₂ (010) layer has an identical arrangement of atoms within the plane; however, every other plane is displaced laterally in the [100] direction by half the length of the unit cell, thereby generating a stacking sequence that repeats in the [010] direction every second layer. The Fe₅C₂ (010) surface termination exposes the repeating (010) plane. This surface is obtained by cleaving the conventional monoclinic unit cell of bulk Fe₅C₂ parallel to the (010) Miller plane at a fractional displacement of 0.25 along the unit-cell axis. The energy cutoff (400 eV) and a (5 × 5 × 1) Monkhorst–Pack grid were used for *k*-point sampling. The adsorption energy (E_{ad}) of CO and K for each reaction was computed according to:

$$E_{\text{ad}} = E_{\text{x/surface}} - E_{\text{surface}} - E_{\text{x}} \quad (1)$$

where $E_{\text{x/surface}}$, E_{x} , and E_{surface} are the total energies of the adsorbed system, the adsorbate ($x = \text{CO}, \text{K}$), and the clean surface, respectively. A negative adsorption energy corresponds to an energetically favorable adsorption site on the surface. Values that are more negative correspond to stronger adsorption. The electronic charge transfer on the atoms was computed using the Bader charge. The variation of electronic charge between the adsorbed and non-adsorbed states of a specific adsorbate was calculated according to:

$$\Delta q_{\text{x}} = q_{\text{x(adsorbed)}} - q_{\text{x(non-adsorbed)}} \quad (2)$$

where q_{x} represents the electronic charge of the adsorbate ($x = \text{CO}, \text{K}$). A negative Bader-charge value indicates that the electrons are transferred from the surface to the adsorbate. The dissociation energy barrier (E_{b}) for each reaction was computed according to:

$$E_{\text{b}} = E_{\text{TS}} - E_{\text{IS}} \quad (3)$$

where E_{b} , E_{TS} , and E_{IS} are the energy barrier of the elementary step, the total energy at the transition state, and the total energy at the initial state, respectively.

The reaction energy (ΔE) for each reaction was computed according to:

$$\Delta E = E_{\text{FS}} - E_{\text{IS}} \quad (4)$$

where E_{TS} , E_{FS} , and E_{IS} are the energy barrier for the elementary step, the total energy at the final state, and the total energy at the initial state, respectively. We employed the climbing image nudged elastic band (CI-NEB) method and constructed the minimum energy path (MEP).

Preparation of the χ -Fe₅C₂/charcoal catalyst

To prepare a K-free χ -Fe₅C₂/charcoal catalyst, 1.85 g of Fe(NO₃)₃·9H₂O was physically ground with 1.0 g of activated charcoal in a mortar for several minutes under ambient conditions until the powder was homogeneously black. Then, the mixed powders were placed in a polypropylene bottle and aged at 323 K in an oven. After aging for 24 h, the sample was cooled in an ambient atmosphere and transferred to an

alumina boat in a tube-type furnace. Finally, the iron-incorporated charcoal powder was slowly heated at a ramping rate of $2.7\text{ }^{\circ}\text{C min}^{-1}$ up to 623 K under a CO flow of 200 mL min^{-1} . The sample was thermally treated at 623 K for 4 h under the continuous CO flow. After the thermal treatment, the resulting black powder was cooled down to room temperature, and then submerged into anhydrous ethanol (20 mL) under N_2 flow in order to prevent rapid surface oxidation. The $\chi\text{-Fe}_5\text{C}_2$ /charcoal powders were simply separated by magnetic force and completely dried in a vacuum oven at 323 K.

Synthesis of the K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal catalyst

For the syntheses of K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal catalysts, stock solutions with different concentrations of aqueous K_2CO_3 solution in distilled water were used (57.2 mM for $\text{K/Fe} = 0.025$, 114.6 mM for $\text{K/Fe} = 0.05$, 171.9 mM for $\text{K/Fe} = 0.075$, 229.2 mM for $\text{K/Fe} = 0.1$). The incipient wetness method was used to merge 1.0 mL of stock solution with the dried $\chi\text{-Fe}_5\text{C}_2$ /charcoal powder. Next, the K-incorporated sample was transferred to an alumina boat in a tube-type furnace, and then slowly heated at a ramping rate of $2.7\text{ }^{\circ}\text{C min}^{-1}$ to 623 K under a CO flow of 200 mL min^{-1} . The sample was calcined with a CO gas flow at the same temperature for 4 h. After the thermal treatment, the resulting black powder was cooled to room temperature, and then was submerged into anhydrous ethanol (20 mL) under N_2 flow. Finally, the K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal powder was separated and dried in a vacuum oven at 323 K.

Catalytic test

The Fischer–Tropsch synthesis reaction was carried out in a fixed-bed reactor composed of stainless steel (5 mm inner diameter and 180 mm length). The catalyst (0.3 g) was diluted with glass beads (3.4 g) to prevent hot spot generation due to a serious exothermic reaction of catalysts and then placed in the fixed-bed reactor. The catalyst was reactivated under a CO flow of 40 mL min^{-1} at 623 K for 4 h. After the activation treatment, the reaction was performed at $320\text{ }^{\circ}\text{C}$ and 15 bar using synthesis gas ($\text{H}_2/\text{CO} = 1.0$, $\text{GHSV} = 8.0\text{ NL g}_{\text{cat}}^{-1}\text{ h}^{-1}$). The composition of the outlet gases was analyzed using an online gas chromatograph (Agilent, 3000A Micro-GC) equipped with a molecular sieve and plot Q columns. The gas flow rates were measured using a wet-gas flow meter (Shinagawa Corp.). After the catalytic reaction, the mixtures of liquid hydrocarbon products and water were isolated in the cold trap at $-3\text{ }^{\circ}\text{C}$, and solid hydrocarbons were obtained in the hot trap at $240\text{ }^{\circ}\text{C}$. The isolated wax and liquid products were analyzed by an offline gas chromatograph (Agilent, 6890 N) with a simulated distillation method (ASTM D2887).

Characterization

The obtained iron carbide/charcoal catalysts were characterized by TEM (Omega EM912 operated at 120 kV, Korea Basic Science Institute), HRTEM (Tecnai G2 F30 operated at 300 kV, KAIST), and an EDX (attached to a F30 Tecnai). For scanning TEM analysis, samples were prepared by putting a few drops of the corresponding colloidal solutions on lacey carbon coated

copper grids (Ted Pella, Inc). High power powder-XRD (Rigaku D/MAX-2500, 18 kW) was also used for analysis. X-ray photoelectron spectroscopy (XPS) studies were carried out using a Sigma Probe (Thermo VG Scientific, Inc.) with a micro-focused monochromator X-ray source. The samples for XPS were prepared by placing a few drops of the colloidal solutions on small pieces ($5\text{ mm} \times 5\text{ mm}$) of gold wafer and then were allowed to dry under N_2 flow. Mössbauer spectra were obtained with a fixed absorber and a moving source. A Mössbauer spectrometer of the electromechanical type, with a 50 mCi ^{57}Co source in a rhodium matrix, was used in constant-acceleration mode. The Fe X-ray absorption spectra (XAS) were recorded on a beam-line 1D of the Pohang Accelerator Laboratory (PAL). N_2 sorption isotherms were measured at 77 K with a TriStar II 3020 surface area analyzer. Before measurement, the samples were degassed in a vacuum at $300\text{ }^{\circ}\text{C}$ for 4 h.

3. Results and discussion

3.1. DFT Calculation for K-doped Fe_5C_2 surfaces

Prior to the catalyst preparation, we modeled a super-cell surface with a slab of five iron layers and three carbon layers. The surface with the “hexagonal” and “square” iron-atom formations was exposed. The K and CO adsorption onto the Fe_5C_2 (010) surface, which consists of a 4-fold hollow, 3-fold hollow, bridge, and top site, were simulated (ESI, Fig. S1†). The first K adsorption, at the K/Fe atomic ratio = 0.025, was simulated by the clean and Fe-terminated surface of Fe_5C_2 (010) containing 40 Fe and 16C atoms for several adsorption sites (Fig. 1a). Among the possible adsorption sites, the K atom preferentially adsorbs on the 3-fold hollow site of Fe because it has the strongest adsorption energy of -2.48 eV (ESI, Table S1†). The Bader charge (q_{K}) at $\text{K/Fe} = 0.025$ was 0.86, which resulted from the electron-charge transfer to the catalyst surface from K. Afterward, another K atom adsorbs onto the nearest stable site, and this process is repeated. All Bader charges for K on the Fe- and C-terminated sites indicated positive values because K donates electrons to the catalyst surface in the role of a Lewis base. The CO adsorption and dissociation properties, after K-doping of the surface, were also affected by nearby K atoms (Fig. 1b). From the calculations of these models, the adsorption energies of CO molecules on the K-doped $\chi\text{-Fe}_5\text{C}_2$ surfaces were obtained (Table 1). As more K atoms spread over the $\chi\text{-Fe}_5\text{C}_2$ surface, the CO adsorption energy and the C–O bond distance ($d_{\text{C-O}}$) increase. A strong CO adsorption was observed simultaneously with the increased C–O bond length. The CO adsorption energies of the K-doped $\chi\text{-Fe}_5\text{C}_2$ surfaces showed -2.26 eV at $\text{K/Fe} = 0.025$, -2.41 eV at $\text{K/Fe} = 0.050$, -2.52 eV at $\text{K/Fe} = 0.075$, and -2.64 eV at $\text{K/Fe} = 0.100$, respectively, indicating the significant K effect on CO adsorption. All Bader charges for CO showed negative values due to the charge transferred from the catalyst surface to CO molecules. Unexpectedly, the energy barrier for CO dissociation and the reaction energy at the K/Fe ratio = 0.075 were lower than those for $\text{K/Fe} = 0.100$, calculated to be 1.40 eV and -1.28 eV , respectively.

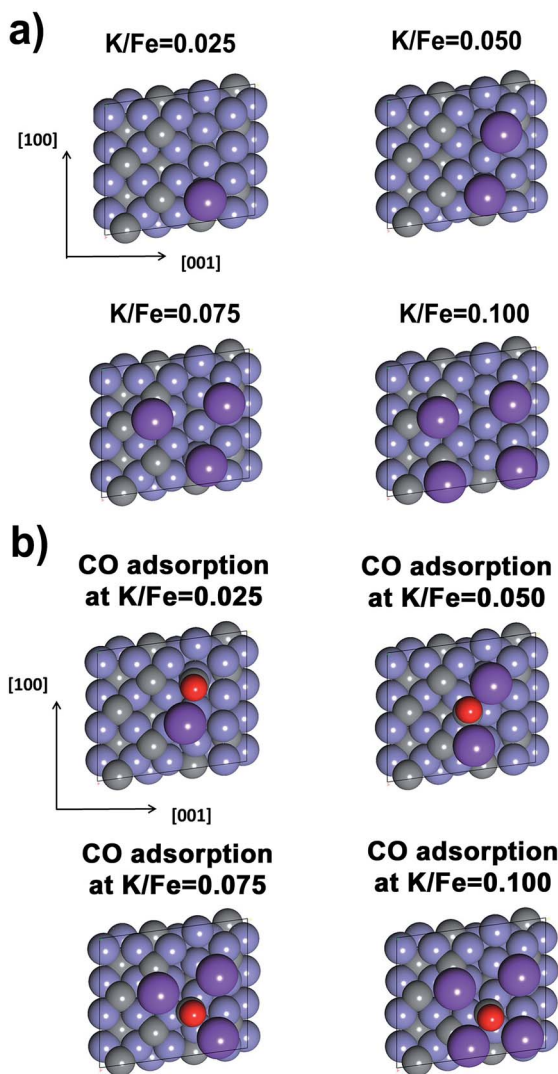
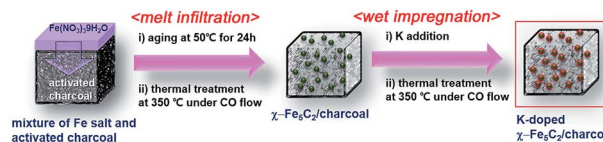


Fig. 1 Surface configurations of (a) stable K adsorption on a clean χ -Fe₅C₂ surface, and (b) stable CO adsorption on a K-doped Fe₅C₂ surface with a controlled K/Fe ratio. The black, blue, red, and violet balls represent carbon, iron, oxygen, and potassium atoms, respectively.

3.2. Synthesis of the K-doped χ -Fe₅C₂/charcoal catalyst

To evaluate the obtained calculation data, we introduced new model catalysts through a melt-infiltration process and an incipient wetness method by controlling the K/Fe atomic ratios (Scheme 1). First, active iron carbide nanoparticles embedded



Scheme 1 Stepwise synthetic scheme of a highly activated K-doped χ -Fe₅C₂/charcoal catalyst.

in a porous charcoal support were successfully obtained *via* two simple steps: an aging step and a thermal decomposition step. In the first step, a Fe(NO₃)₃·9H₂O salt was homogeneously mixed with activated charcoal by grinding at room temperature. This mixture was subsequently melt-infiltrated into mesoporous charcoal by aging at 323 K for 24 h in an oven. This melt-infiltration process, which is also well known as the solvent-free method or solid-liquid route, has already been employed for the preparation of some supported catalysts due to its remarkable convenience and speed.^{47,48} In the second step, the melt-infiltrated Fe salt was transformed to active iron carbide nanoparticles by thermal decomposition at 623 K under a flow of CO.

The transmission electron microscopy (TEM) image shows the incorporated iron carbide nanoparticles with an average diameter of 17.8 ± 2.8 nm (Fig. 2a and i). In the high-resolution TEM (HRTEM) image, the iron carbide particle in the porous charcoal shows its single crystalline nature and the lattice distance was measured to be 0.205 nm, consistent with the value of the Fe₅C₂ (510) plane (Fig. 2b). Next, K-doped Fe₅C₂ particles were obtained by impregnating the K₂CO₃ salt as a K source into the Fe₅C₂/charcoal powders and thermally treating them. The K/Fe ratios in the K-doped Fe₅C₂ particles were controlled using different concentrations of aqueous K₂CO₃ solution. The K-doped Fe₅C₂ nanoparticles at the K/Fe ratio = 0.075 were well dispersed in the charcoal with their relatively narrow size distribution (18.0 ± 2.6 nm) remaining without any significant particle agglomerations even after the high-temperature thermal retreatment of the Fe₅C₂ particles at 623 K (Fig. 2c and j). The HRTEM image and the corresponding Fourier-transform (FT) pattern show that the final K-doped Fe₅C₂ particle with a spherical shape was a single crystal with the distance of 0.205 nm between neighboring fringes (Fig. 2d). High angle annular dark field-scanning transmission electron microscopy (HAADF-STEM) and elemental mapping of carbon (red), iron (green), and potassium (brown) clearly showed the uniform distribution of each component (Fig. 2e-h). While

Table 1 Calculated adsorption energies (E_{ad}), distances for CO, and energy barriers (E_b) and reaction energies (ΔE) for CO dissociation on Fe₅C₂ surfaces

K/Fe atomic ratio	E_{ad} (eV) of CO	d_{C-O} (Å)	$q_{CO}(\Delta e)$	E_b (eV) for CO dissociation	ΔE (eV)
K-free	-2.08	1.211	-0.86	1.74	-1.12
K/Fe = 0.025	-2.26	1.224	-0.96	1.64	-1.16
K/Fe = 0.050	-2.41	1.248	-1.09	1.53	-1.21
K/Fe = 0.075	-2.52	1.262	-1.19	1.40	-1.28
K/Fe = 0.100	-2.64	1.275	-1.38	1.48	-1.22

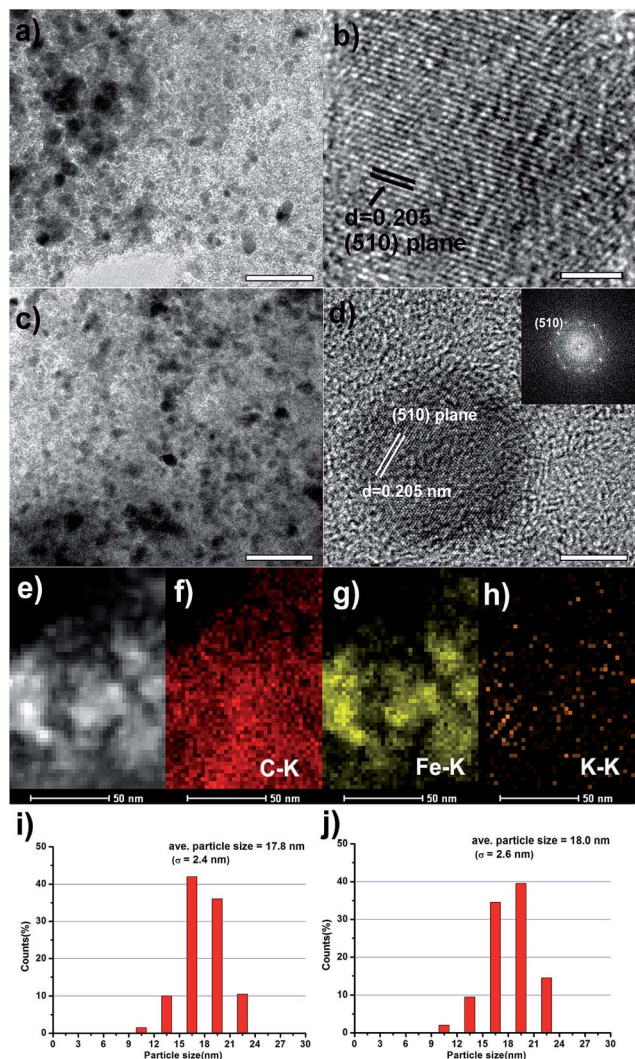


Fig. 2 TEM images of (a) K-free and (c) K-doped χ -Fe₅C₂ (K/Fe = 0.075) nanoparticles on activated charcoal, and HRTEM images of (b) K-free and (d) K-doped χ -Fe₅C₂ single particle with the corresponding FT pattern (inset of d). (e) HAADF-STEM image and (f–h) elemental mapping for the K-doped χ -Fe₅C₂/charcoal catalyst (K/Fe = 0.075). Particle size distribution histograms of (i) K-free and (j) K-doped χ -Fe₅C₂ nanoparticles (K/Fe = 0.075). The bars represent 100 nm (a and c), 2 nm (b), and 5 nm (d), respectively.

carbon could be detected over the whole area, potassium was mainly found in the Fe-rich area, meaning that the added potassium atoms were well adsorbed onto the Fe₅C₂ particles in charcoal. All the K-doped Fe₅C₂ particles even with different K/Fe ratios showed similar average particle sizes, measured to be 17.8 ± 2.7 nm at K/Fe = 0.025, 18.2 ± 2.8 nm at K/Fe = 0.050, and 20.4 ± 3.9 nm at K/Fe = 0.100 (Fig. S2†). In Fig. 3a, the X-ray diffraction (XRD) spectra show that all iron carbide particles are well-matched with the two Hägg-carbide phases (JCPDS no. 36-1248 and no. 51-0997; space group, *C2/c*). No significant differences in the XRD spectra for all samples were observed. A sharp peak at $2\theta = 26.5^\circ$ corresponded to a crystalline carbon structure in the activated charcoal (JCPDS no. 26-1077). The crystallite sizes of K-free and K-doped Fe₅C₂ estimated from the

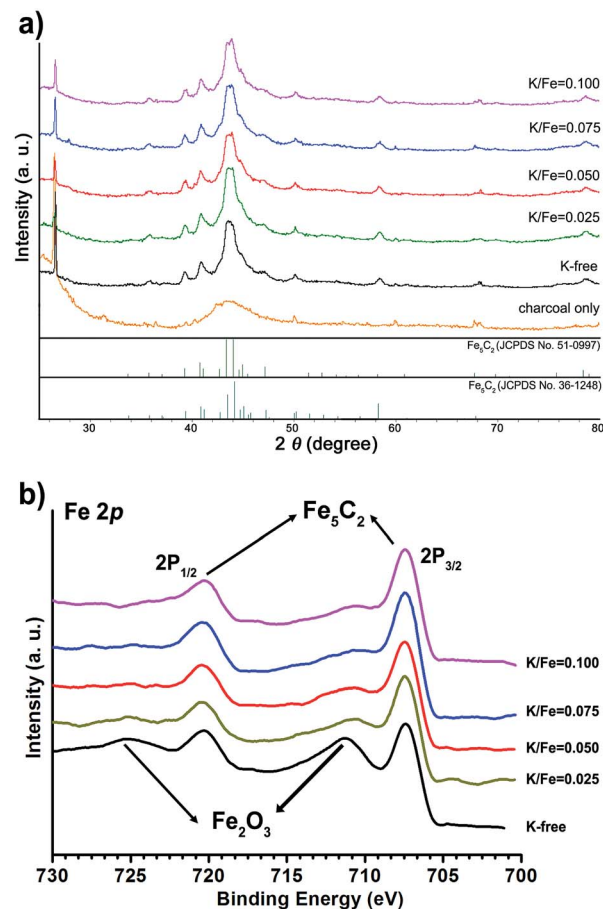


Fig. 3 (a) XRD and (b) XPS spectra of K-free and K-doped χ -Fe₅C₂ nanoparticles on activated charcoal.

FWHM of the (020) diffraction peaks were 20.0 nm at K-free, 20.8 nm at K/Fe = 0.025, 20.8 nm at K/Fe = 0.050, 19.9 nm at K/Fe = 0.075, and 21.0 nm at K/Fe = 0.100. The calculated single crystallite sizes were very close to the average particle sizes measured in the TEM images, demonstrating the single crystal nature. To investigate the surface states in the K-free and K-doped Fe₅C₂ particles, the core-level X-ray photoelectron spectroscopy (XPS) spectrum of Fe for each sample was measured (Fig. 3b). All XPS spectra show two sets of peaks from iron carbide and iron oxide in the range of Fe 2p_{3/2} and 2p_{1/2} orbital signals. In the spectra, the surfaces of the Fe₅C₂ nanoparticles, for both the K-free and K-doped samples, were observed as partially oxidized forms. Even though the fresh catalyst powders were passivated by ethanol, the slight surface oxidation of the highly active Fe₅C₂ nanoparticles was inevitable due to atmospheric exposure during the sampling processes. Interestingly, all K-doped Fe₅C₂ nanoparticles showed much better stability against oxidation than did K-free Fe₅C₂ nanoparticles, because the neighbor K could be more easily oxidized than Fe₅C₂. Mössbauer investigation has been applied to measure the rate of carbiding of iron during the FTS reaction, over a supported iron catalyst.^{49,50} The specific portion of the pure Hägg-carbide species in several iron carbide phases could be ascertained by Mössbauer spectroscopy. To investigate the

distribution of the iron based emergent materials and hyperfine interaction, the Mössbauer spectra of the K-free $\chi\text{-Fe}_5\text{C}_2$ /charcoal were obtained at 4.2 and 295 K. The Mössbauer spectra show the superposition of three-sextets for Fe sites, corresponding to only $\chi\text{-Fe}_5\text{C}_2$ crystallography sites. Other iron carbide phases, such as $\text{Fe}_{2.2}\text{C}$, Fe_7C_3 , Fe_3C , were not observed. The one-doublet at 295 K is analyzed to be a Fe_2O_3 phase (Fig. 4). The relative areas of $\chi\text{-Fe}_5\text{C}_2$ and Fe_2O_3 , which were obtained from the fitted Mössbauer parameters, were calculated to be 88% and 12%, respectively (Table S2†).

To check the oxidation state in the bulk form and binding environment for Fe atoms within C molecules for the K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal, an X-ray absorption spectroscopy (XAS) measurement was conducted. This is based on electron excitation by an X-ray beam which moves an electron from a core orbital to an unoccupied molecular orbital. In the X-ray absorption near-edge structure (XANES) region from an absorption edge to 7162 eV, Fe pre-edge peaks for all K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal were observed at 7112 eV, which corresponds with that of metallic Fe (Fig. S3a and b†). All the extended X-ray absorption fine structure (EXAFS) patterns for K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal showed two major peaks of the first two coordination shells of Fe_5C_2 , originating from Fe–C (~ 1.6 Å) and Fe–Fe (~ 2.2 Å) scattering (Fig. S3c and d†).

The pore volume and size of the porous charcoal support were crucial to determine the loading ratio of iron salt by the

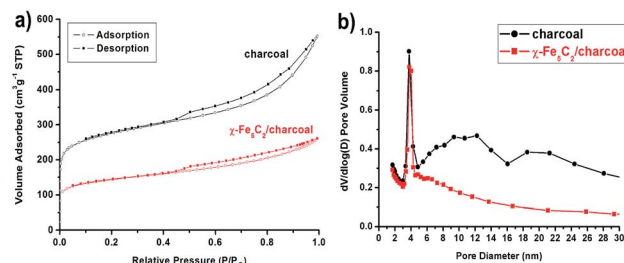


Fig. 5 (a) N_2 adsorption–desorption isotherms of the initial charcoal and K-doped $\chi\text{-Fe}_5\text{C}_2$ on activated charcoal (K/Fe = 0.075), (b) pore size distribution diagrams of K-doped $\chi\text{-Fe}_5\text{C}_2$ on activated charcoal (K/Fe = 0.075) calculated from desorption branches using the BJH method.

melt-infiltration method. N_2 sorption experiments at 77 K for the K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal catalyst at K/Fe = 0.075 exhibited type IV adsorption–desorption hysteresis with delayed capillary evaporation at a relative pressure of 0.5 (Fig. 5a). The Brunauer–Emmett–Teller (BET) surface areas of the initial charcoal and the K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal were calculated to be $1010\text{ m}^2\text{ g}^{-1}$ and $525\text{ m}^2\text{ g}^{-1}$, respectively. The total pore volumes were also measured to be $0.85\text{ cm}^3\text{ g}^{-1}$ for the initial charcoal and $0.40\text{ cm}^3\text{ g}^{-1}$ for the K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal. The small pore sizes of the initial charcoal and the K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal were found to be equal (3.8 nm) using the Barrett–Joyner–Halenda (BJH) method from their desorption branches (Fig. 5b).

High Fe-loading with uniform particle dispersion in the supported Fe catalyst is also very important. The Fe-loading content for the K-free and K-doped catalysts was calculated to be approximately 20 wt% on the basis of Fe converted from the iron nitrate salt after the thermal treatment.

3.3. High-temperature Fischer–Tropsch synthesis reaction

High-temperature FTS reaction tests were carried out at 15 bar, 320°C , and an H_2/CO ratio of 1 using the model catalysts. The

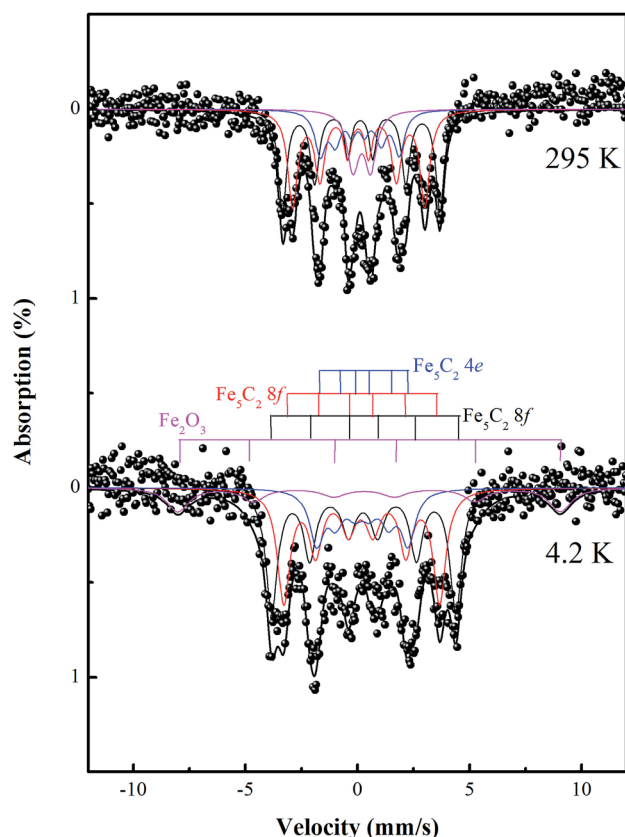


Fig. 4 Mössbauer spectra of K-free $\chi\text{-Fe}_5\text{C}_2$ nanoparticles on activated charcoal at 4.2 and 295 K.

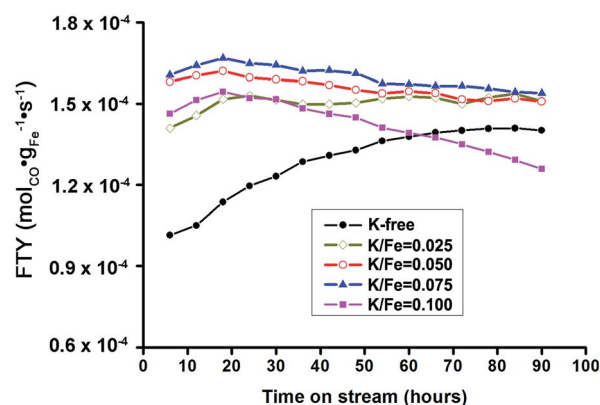


Fig. 6 Catalytic performance of K-free and K-doped $\chi\text{-Fe}_5\text{C}_2$ /charcoal catalysts for high-temperature FTS. The reaction tests were conducted at 320°C and at 15 bar and a $\text{H}_2:\text{CO}$ ratio of 1. FTY represents the number of CO moles converted to hydrocarbons per gram of iron per second.

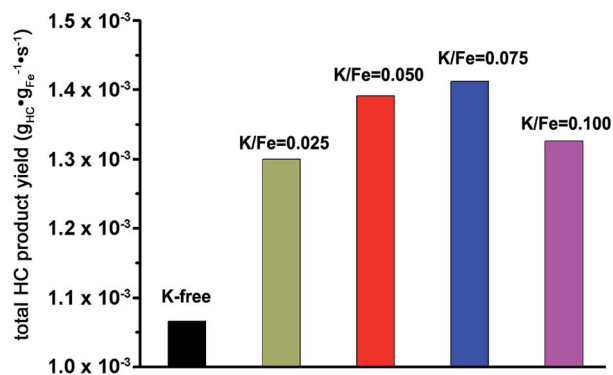


Fig. 7 Total hydrocarbon product yield for K-free and K-doped χ - Fe_5C_2 /charcoal catalysts after 90 h on stream.

catalytic activity was noted as the iron-time-yield (FTY, *i.e.*, the number of CO moles converted to hydrocarbons per gram of iron per second) over time-on-stream (TOS). These FTY values reflect the CO conversion and hydrocarbon selectivity of each FTS catalyst. As expected from the calculation results, the K-doped catalysts showed much better catalytic performance, with higher FTY values, than that of the K-free catalyst (Fig. 6). In addition, the initial activity for 48 h in all K-doped Fe_5C_2 /charcoal catalysts was superior to that of the K-free catalyst, thus demonstrating their high and fast activation with K. The relatively long induction period during FTS over the K-free catalyst is mainly due to the more difficult phase transformation from the partially oxidized iron to activated Fe_5C_2 species. In particular, the K-doped catalyst at $\text{K/Fe} = 0.075$ showed a very high CO conversion (>95%) with a good C_{5+} selectivity ($\sim 50\%$) (Fig. S4d and S5d[†]). Moreover, the selectivity for methane as a side product was also very low ($\sim 5\%$), comparable to 12% of the K-free catalyst (Fig. S5[†]). Based on these results, at $\text{K/Fe} = 0.075$, the highest activity ($1.54 \times 10^{-4} \text{ mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$) was obtained for up to 90 h. This FTY value is much higher than that of the previously reported highest value ($8.48 \times 10^{-5} \text{ mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$) in various supported iron nanostructures.¹⁶ In the previous works, K-doped Fe on carbon spheres and K-doped Fe on carbon nanotubes showed relatively low activities of $1.8 \times 10^{-5} \text{ mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$ and $2.8 \times 10^{-5} \text{ mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$, respectively.^{11,15} We believe that the enhanced activity mainly originates from the electrostatic interaction by the added K,^{51,52} not from any changes of active facets, because significant differences among the catalyst surface structures and crystallite sizes were not detected in the XRD data. On the

other hand, the activity at $\text{K/Fe} = 0.1$ gradually decreased to $1.26 \times 10^{-4} \text{ mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$ under the same reaction conditions. At $\text{K/Fe} = 0.1$, the deactivation with low FTY values after 20 h is closely linked to the increased CO_2 selectivity and decreased C_{5+} selectivity (Fig. S5e[†]). As seen in the DFT calculation results of Table 1, the catalyst at $\text{K/Fe} = 0.1$ showed an unfavorable reaction energy (-1.22 eV) for CO dissociation than -1.28 eV of $\text{K/Fe} = 0.075$, even though it had the strongest CO adsorption energy (-2.64 eV). Because of the steric effect of CO caused by a large amount of neighboring K, the CO dissociation energy barrier (1.48 eV) at $\text{K/Fe} = 0.1$ was higher than 1.40 eV at $\text{K/Fe} = 0.075$.

The total hydrocarbon (HC) product yields (gram of generated hydrocarbons per gram of iron per second) for the K-free and K-doped χ - Fe_5C_2 catalysts were measured after 90 h on stream of the reaction, calculated by the sum of specific product yields (Fig. 7 and Table 2). The specific product yield for each sample was obtained by the gas chromatography (GC) analysis of gaseous products (C_1 – C_4) and simulated distillation (SIMDIS) analysis of isolated solid (wax) and liquid (oil) products. In all K-doped Fe_5C_2 catalysts, total HC product yields higher than that of the K-free χ - Fe_5C_2 catalyst were obtained. Among the K-doped χ - Fe_5C_2 catalysts, the total HC product yield at $\text{K/Fe} = 0.075$ was the best, calculated to be $1.41 \times 10^{-3} \text{ g}_{\text{HC}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$. In contrast, the K-free catalyst showed a very low total HC productivity ($1.07 \times 10^{-3} \text{ g}_{\text{HC}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$). In the specific product yields, the K-addition effect was observed to be dramatic. In all the K-doped χ - Fe_5C_2 catalysts, the methane product was significantly decreased compared to that of K-free χ - Fe_5C_2 . In particular, regarding the production of gasoline range products (C_5 – C_{12}), the K-doped χ - Fe_5C_2 catalyst at $\text{K/Fe} = 0.075$ showed a $4.86 \times 10^{-4} \text{ g}_{\text{HC}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$ product yield, which was 8.5 times higher than that of the K-free χ - Fe_5C_2 catalyst. For the product analysis, the hydrocarbon distributions were also measured after 90 h (Table S3[†]). The distributions for lower olefins and gasoline at $\text{K/Fe} = 0.075$ were $\sim 24 \text{ wt\%}$ and $\sim 34 \text{ wt\%}$ in the total hydrocarbon products, respectively.

4. Conclusions

We have conducted theoretical and experimental research on K-free and K-doped χ - Fe_5C_2 on charcoal nanocatalysts. The DFT calculation of the K doping effects on the active Hägg-carbide phase suggested that the optimum K-doping ratio is $\text{K/Fe} = 0.075$, consistent with the experiment results. The K-doped

Table 2 Specific product yields for K-free and K-doped χ - Fe_5C_2 /charcoal catalysts after 90 h on stream

Cat.	Specific product yield ($10^{-4} \text{ g}_{\text{HC}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$)					
	CH_4	C_2 – C_4 olefins	C_2 – C_4 paraffins	C_5 – C_{12}	C_{13} – C_{18}	C_{19+}
K-free	4.11	1.08	4.67	0.57	0.15	0.09
K/Fe = 0.025	3.10	2.05	3.47	3.63	0.53	0.26
K/Fe = 0.050	1.60	3.44	1.24	4.80	1.44	1.38
K/Fe = 0.075	1.50	3.36	1.10	4.86	1.64	1.67
K/Fe = 0.100	1.20	3.07	0.66	4.50	1.73	2.11

χ -Fe₅C₂/charcoal nanocatalysts with a controlled K/Fe atomic ratio, were prepared *via* a facile melt-infiltration process and incipient wetness method, showed the highest FTY values with the best total HC product yield in high-temperature FTS under 320 °C and at 15 bar. It is anticipated that this approach based on computational simulation will be useful in creating more delicate and rational designs of heterogeneous catalysts by improving our understanding of some additive effects.

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Notes and references

- Q. Zhang, J. Kang and Y. Wang, *ChemCatChem*, 2010, **2**, 1030.
- R. Lique, A. R. Campelo, J. M. Campelo, A. A. Romero, J. L. Valverde and P. Sanchez, *Energy Environ. Sci.*, 2012, **5**, 5186.
- M. E. Dry, *Catal. Today*, 2002, **71**, 227.
- Studies in Surface Science and Catalysis: Fischer-Tropsch Technology*, ed. A. P. Steynberg and M. E. Dry, Elsevier, Amsterdam, 2004, vol. 152.
- A. de Klerk, *Energy Environ. Sci.*, 2011, **4**, 1177.
- A. de Klerk, *Green Chem.*, 2007, **9**, 560.
- A. P. Steynberg, R. L. Espinoza, B. Jager and A. C. Vosloo, *Appl. Catal., A*, 1999, **186**, 41.
- H.-J. Wan, B.-S. Wu, C.-H. Zhang, H.-W. Xiang, Y.-W. Li, B.-F. Xu and F. Yi, *Catal. Commun.*, 2007, **8**, 1538.
- C.-H. Zhang, H.-J. Wang, Y. Yang, H.-W. Xiang and Y.-W. Li, *Catal. Commun.*, 2006, **7**, 733.
- J. Shen, B. Guang, M. Tu and Y. Chen, *Catal. Today*, 1996, **30**, 77.
- H. Xiong, M. Moyo, M. A. M. Motchelaho, L. L. Jewell and N. Coville, *Appl. Catal., A*, 2010, **388**, 168.
- R. M. Abbaslou, A. Tavassoli, J. Soltan and A. K. Dalai, *Appl. Catal., A*, 2009, **367**, 47.
- J. F. Bengoa, A. M. Alvarez, M. V. Cagnoli, N. G. Gallegos and S. G. Marchetti, *Appl. Catal., A*, 2007, **325**, 68.
- G. Yu, B. Sun, Y. Pei, S. Xie, S. Yan, M. Qiao, K. Fan, X. Zhang and B. Zong, *J. Am. Chem. Soc.*, 2010, **132**, 935.
- J. Lu, L. Yang, B. Xu, Q. Wu, D. Zhang, S. Yuan, Y. Zhai, X. Wang, Y. Fan and Z. Hu, *ACS Catal.*, 2014, **4**, 613.
- H. M. T. Galvis, J. H. Bitter, C. B. Khare, M. Ruitenbeek, A. I. Dugulan and K. P. de Jong, *Science*, 2012, **335**, 835.
- H. M. T. Galvis, J. H. Bitter, T. Davidian, M. Ruitenbeek, A. I. Dugulan and K. P. de Jong, *J. Am. Chem. Soc.*, 2012, **134**, 16207.
- E. de Smit and B. M. Weckhuysen, *Chem. Soc. Rev.*, 2008, **37**, 2758.
- D. B. Bukur, K. Okabe, M. P. Rosynek, C. Li, D. Wang, K. R. P. M. Rao and G. P. Huffman, *J. Catal.*, 1995, **155**, 353.
- A. Sarkar, D. Seth, A. K. Dozier, J. K. Neathery, H. H. Hamdeh and B. H. Davis, *Catal. Lett.*, 2007, **117**, 1.
- E. de Smit, M. M. van Schooneveld, F. Cinquini, H. Bluhm, P. Sautet, F. M. F. de Groot and B. M. Weckhuysen, *Angew. Chem., Int. Ed.*, 2011, **50**, 1584.
- T. Herranz, S. Rojas, F. J. Pérez-Alonso, M. Ojeda, P. Terreos and J. L. Fierro, *J. Catal.*, 2006, **243**, 199–211.
- E. de Smit, F. Cinquini, A. M. Beale, O. V. Safonova, W. van Beek, P. Sautet and B. M. Weckhuysen, *J. Am. Chem. Soc.*, 2010, **132**, 14928.
- C. Yang, H. Zhao, Y. Hou and D. Ma, *J. Am. Chem. Soc.*, 2012, **134**, 15814.
- R. J. O'Brien, L. Xu, D. R. Milburn, Y.-X. Li, K. J. Klabunde and B. H. Davis, *Top. Catal.*, 1995, **2**, 1.
- G. Zhao, C. Zhang, S. Qin, H. Xiang and Y. Li, *J. Mol. Catal. A: Chem.*, 2008, **286**, 137.
- S. Li, S. Krishnamoorthy, A. Li, G. D. Meitzner and E. Iglesia, *J. Catal.*, 2002, **206**, 202.
- N. Lohitharn, J. G. Goodwin Jr and E. Lotero, *J. Catal.*, 2008, **255**, 104.
- A. N. Pour, M. R. Housaindokht, S. F. Tayyari, J. Zarkesh and M. R. Alaei, *J. Nat. Gas Sci. Eng.*, 2010, **2**, 61.
- W. Ngantsoue-Hoc, Y. Zhang, R. J. O'Brien, M. Luo and B. H. Davis, *Appl. Catal., A*, 2002, **236**, 77.
- S. Li, A. Li, S. Krishnamoorthy and E. Iglesia, *Catal. Lett.*, 2001, **4**, 197.
- B. Graf and M. Muhler, *Phys. Chem. Chem. Phys.*, 2011, **13**, 3701.
- Y. Yang, H.-X. Xiang, Y.-Y. Xu, L. Bai and Y.-W. Li, *Appl. Catal., A*, 2004, **266**, 181.
- C. Harding, V. Habibpour, S. Kunz, A. N.-S. Farnbacher, U. Heiz, B. Yoon and U. Landman, *J. Am. Chem. Soc.*, 2009, **131**, 538.
- J. K. Nørskov, T. Bligaard, J. Rossmeisl and C. H. Christensen, *Nat. Chem.*, 2009, **1**, 37.
- X.-Q. Zhang, E. Iype, S. V. Nedeia, A. P. J. Jansen, B. M. Szyja, E. J. M. Hensen and R. A. van Santen, *J. Phys. Chem. C*, 2014, **118**, 6822.
- K. Honkala, A. Hellman, I. N. Remediakis, A. Logadottir, A. Carlsson, S. Dahl, C. H. Christensen and J. K. Nørskov, *Science*, 2005, **307**, 555.
- F. Studt, F. Abild-Pedersen, T. Bligaard, R. Z. Sørensen, C. H. Christensen and J. K. Nørskov, *Science*, 2008, **320**, 1320.
- D.-B. Cao, Y.-W. Li, J. Wang and H. Jiao, *J. Mol. Catal. A: Chem.*, 2011, **346**, 55.
- S. Zhao, X.-W. Liu, C.-F. Huo, Y.-W. Li, J. Wang and H. Jiao, *J. Catal.*, 2012, **294**, 47.
- C.-F. Huo, Y.-W. Li, J. Wang and H. Jiao, *J. Am. Chem. Soc.*, 2009, **131**, 14713.
- D. C. Sorescu, *Surf. Sci.*, 2011, **605**, 401.
- D. C. Sorescu, *J. Phys. Chem. C*, 2009, **113**, 9256.
- C.-F. Huo, B.-S. Wu, P. Gao, Y. Yang, Y.-W. Li and H. Jiao, *Angew. Chem., Int. Ed.*, 2011, **50**, 7403.
- G. Kresse and D. Joubert, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1999, **59**, 1758.

- 46 M. A. Petersen, J.-A. van den Berg and W. J. van Rensburg, *J. Phys. Chem. C*, 2010, **114**, 7863.
- 47 P. E. de Jongh, R. W. P. Wagemans, T. M. Eggenhuisen, B. S. Dauvillier, P. B. Radstake, J. D. Meeldijk, J. W. Geus and K. P. de Jong, *Chem. Mater.*, 2007, **19**, 6052.
- 48 T. M. Eggenhuisen, J. P. den Breejen, D. Verdoes, P. E. de Jongh and K. P. de Jong, *J. Am. Chem. Soc.*, 2010, **132**, 18318.
- 49 G. B. Raupp and W. N. Delgass, *J. Catal.*, 1979, **58**, 361.
- 50 J. W. Niemantsverdriet, A. M. Van der Kraan, W. L. Van Dijk and H. S. Van der Baan, *J. Phys. Chem.*, 1980, **84**, 3363.
- 51 Z.-P. Liu and P. Hu, *J. Am. Chem. Soc.*, 2001, **123**, 12596.
- 52 R. W. Gurney, *Phys. Rev.*, 1935, **47**, 479.